



Kirby Lau <kirbylau0@gmail.com>

Re: FLL Innovation Project Inquiry: Stouffville Road Jefferson Salamander Wildlife Tunnels (Team #65536)

1 message

Hugh Guan <support@joyrobots.org>

To: Jonathan Ruppert <Jonathan.Ruppert@trca.ca>

Cc: David Lawrie <David.Lawrie@trca.ca>, Karen McDonald <Karen.McDonald@trca.ca>, Info <Info@trca.ca>, kirbylau0@gmail.com

Dear Dr. Ruppert,

Thank you so much from all of us!

When I shared your reply with our team during our weekend session, the kids were thrilled to learn that you and David have led the road ecology program at Stouffville Road for over a decade.

Your closing advice—"build, test, learn, and improve"—struck a huge chord with our 9- and 10-year-olds. The team wrote it at the very top of their engineering whiteboard as our official team motto.

Learning that monitoring tunnel completion is an active, ongoing effort gave our young team clear engineering focus:

- To explore non-invasive ways to monitor dark, wet tunnels without flash or lens fouling, the students chose to experiment with an overhead **Time-of-Flight (ToF) laser distance array** and microphone.
- They are currently assembling a **1:5 scale acrylic test sandbox of the DECAST culvert** to test whether multi-beam sequencing can tell if a salamander completes its crossing or turns back.

A direct request from the students:As we complete the physical prototype over the next 3 to 4 weeks, the kids asked if there might be an opportunity for a **brief 10- to 15-minute virtual meeting (Teams / Zoom)** with you or a member of your team.

They would love to give you a quick 2-minute live demo of their sandbox and ask 2 or 3 quick technical questions about culvert conditions.

(We know how demanding your schedule is—if an online call is not feasible, the kids would be just as grateful to email you a 2-minute recorded demo video and a 1-page drawing for a few bullet points.)

Thank you again for inspiring our students and for your dedication to protecting Richmond Hill's local ecosystems!

Warm regards,

Hugh GuanOn behalf of the students of **JoyRobots Excavators (Team #65536)**Richmond Hill, ON | 6474486124 | joyrobots.org

On Oct 1, 2026, at 12:54 PM, Jonathan Ruppert <Jonathan.Ruppert@trca.ca> wrote:

Hi JoyRobots Excavators (Team #65536),

Thanks for reaching out. We'd be happy to answer your questions. David and I lead TRCA's road ecology program and have worked at the Stouffville Road crossing for many years. Please feel free to contact us if you have any follow-up questions.

Why are you so focused on Jefferson Salamanders?

Jefferson Salamanders are endangered in Ontario and need both forests and ponds to complete their life cycle. At Stouffville Road, they must travel between crossing the road. TRCA has studied salamanders in this area since 2002 to better understand their movements and how best to protect them.

About how many Jefferson Salamanders die each year?

We don't know the exact number, so we don't want to guess. Past road mortality surveys show that salamanders are killed by vehicles in this area. They are often found on cool, rainy nights. Our research identified this section of Stouffville Road as an important location for reducing that risk.

Around how many Jefferson Salamanders use the tunnels?

We don't have a reliable number yet. The tunnels were installed in 2025, so monitoring is still underway. TRCA scientists are using specialized cameras to see if they cross, and whether Jefferson Salamanders are using them.

What makes salamanders comfortable in the tunnels?

Jefferson Salamanders prefer cool, dark, damp conditions and spend much of their lives underground in animal burrows. Tunnel entrances also need to be helped guide salamanders toward the tunnels and away from traffic.

What else do you do to help salamanders?

Tunnels are only one part of the solution. We study where salamanders live, breed, overwinter, and travel. We also use tools such as eDNA to look for additional signs of salamanders, create new breeding ponds, and work with York Region on seasonal road closures during salamander migrations.

Are the tunnels under Stouffville Road working?

The best answer right now is: we're still finding out. Wildlife has been detected using the new tunnels, which is encouraging, but we are still studying how w Salamanders. We are also checking the tunnels and fencing for problems and identifying ways to improve them. That is an important part of science: build,

Regards,

Jonathan

Jonathan Ruppert, PhD (he, him)

Senior Manager, Ecosystem & Climate Science
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From: Karen McDonald <Karen.McDonald@trca.ca>

Sent: Tuesday, September 29, 2026 10:39

To: David Lawrie <David.Lawrie@trca.ca>; Jonathan Ruppert <Jonathan.Ruppert@trca.ca>

Subject: FW: FLL Innovation Project Inquiry: Stouffville Road Jefferson Salamander Wildlife Tunnels (Team #65536)

Hey Jonathan and Dave,

Here's an inquiry for you 😊

Best,
Karen

From: Info <Info@trca.ca>

Sent: September 29, 2026 10:10 AM

To: Karen McDonald <Karen.McDonald@trca.ca>

Subject: FW: FLL Innovation Project Inquiry: Stouffville Road Jefferson Salamander Wildlife Tunnels (Team #65536)

Hello,

Please see inquiry below or redirect to the appropriate staff for review/action and respond directly to the sender.

Your assistance to this matter is greatly appreciated.

Thank you and have a great day!

Kathy Papadopoulos

Receptionist
Property and Asset Management | Legislative and Property Services

T: (416) 661-6600

E: kathy.papadopoulos@trca.ca

A: [5 Shoreham Drive, Toronto, ON, M3N 1S4](#) | trca.ca



Toronto and Region Conservation Authority



From: Hugh Guan <support@joyrobots.org>

Sent: Saturday, September 26, 2026 6:17 PM

To: Info <info@trca.ca>; education@trca.ca

Subject: FLL Innovation Project Inquiry: Stouffville Road Jefferson Salamander Wildlife Tunnels (Team #65536)

EXTERNAL SENDER

Dear TRCA Environmental Monitoring Team,

We are the **JoyRobots Excavators (#65536)**, a youth FIRST LEGO League (FLL) robotics team of 9- and 10-year-olds based in Richmond Hill. For this season's Innovation Project, we are working on a wildlife underpasses along Stouffville Road designed to protect the endangered Jefferson Salamander.

As we work on an engineering concept to study non-invasive tunnel monitoring, we would love to learn from your real-world research and field experience.

Our team brainstormed these 6 questions for your experts:

1. Why are you so centred in Jefferson Salamanders?
2. About how many Jefferson salamanders die each year?
3. Around how many Jefferson Salamanders use these tunnel crossings?
4. What will make the salamanders feel comfortable in the tunnels?
5. What do you do other than making tunnels to help salamanders?
6. Is the tunnel working under stouffville road?

Thank you very much for your time and for everything you do to protect our local wildlife and ecosystems!

Sincerely,

JoyRobots Excavators (Team #65536)

Richmond Hill, ON

Coach Contact: Hugh Guan / support@joyrobots.org / 6474486124